

ORIGINAL ARTICLE



Evaluating the effectiveness of the Complex Behaviour Forum—A multisystem approach to supporting people with complex challenging behaviour

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Abstract

Background: This study assessed the effectiveness of a cross-departmental case review panel—the Multicap Complex Behaviour Forum (CBF)—in reducing challenging behaviour exhibited by people with intellectual disabilities.

Methods: Thirty participants (15 CBF participants and 15 matched-control participants) took part in the study. Behavioural data was collected for each CBF participant (and their matched control) for the three-month period before entering the CBF, during their time in the CBF, and the 3 months after exiting the CBF.

Results: There was a significant interaction of group and time observed, with the CBF participants showing more change in behavioural incidents across time. Associated with this change was a noticeable reduction in staff injury costs related to the challenging behaviour of the CBF participants.

Conclusions: This study demonstrates that positive behavioural and organisational outcomes are enhanced by fostering collaboration across multiple organisational systems when it comes to supporting people who exhibit challenging behaviours.

KEYWORDS

challenging behaviour, clinical governance, positive behaviour support, restrictive practice

1 | INTRODUCTION

Emerson and Einfeld (2011) define challenging behaviour as culturally abnormal behaviour that is likely to result in harm to the individual or others, or that will result in them being excluded from the broader community. The impact of challenging behaviours engaged in by individuals with intellectual disability, like aggression, self-injury, and destructiveness, are well documented (Ali et al., 2018; Emerson & Einfeld, 2011; Lowe et al., 2007). There are a variety of ways to effectively support individuals who exhibit such behaviours, and may

include environmental management (Emerson & Einfeld, 2011), chemical interventions (e.g., psychiatric medications) (Emerson & Einfeld, 2011), and behavioural approaches such as applied behaviour analysis (ABA) and positive behaviour support (PBS) (Carr et al., 2016; Tincani, 2007). PBS as a discipline has evolved from the behaviour change model of ABA (Carr et al., 2016) and in addition to ABA concepts, PBS emphasises person-centredness, human rights, quality of life, normalisation, participation, and inclusion (Carr et al., 2016; Johnston et al., 2006). Given the significant impact of challenging behaviours, and the prevalence of people with an intellectual disability

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who engage in them, addressing such behaviour necessitates a carefully considered clinical response (Lowe et al., 2007). This paper outlines the work of Multicap, an Australian disability service provider, and its efforts to establish a specialised group to plan interventions for individuals engaging in complex challenging behaviours (behaviours which have been resistant to the standard level of PBS intervention) who require expert oversight in their PBS strategies.

Multicap is a disability support organisation with a large footprint throughout Queensland, Northern New South Wales and Victoria, and offers a wide range of supports such as behaviour support, supported accommodation, community and in-home support, employment services, school holiday programs, and case management. Multicap supports over 1700 people with a disability and is a leading provider for people with high support needs. Approximately 10% of Multicap's total clients receive targeted behaviour support that involves a behaviour support clinician. This figure approaches 40% when considering clients who also access Multicap's supported and independent accommodation services. These ratios are broadly consistent with overseas research on the prevalence of challenging behaviour engaged in by people with intellectual disability (Emerson, 2001; Poppes et al., 2010).

The Multicap PBS Team has been structured based on recent PBS practice reviews, particularly those emerging from Australia (Carmichael et al., 2020; Hayward et al., 2019; Wardale et al., 2018). Led by a Board-Certified Behaviour Analyst (BCBA), the team comprises 40 clinicians recruited from post-graduate fields like ABA, psychology, and social work (Behaviour Analyst Certification Board, 2023). Multicap offers a structured program of supporting postgraduate education through both financial reimbursements, and the provision of internal clinical supervisors. Any new clinician who has not studied ABA at Masters level is required to complete a short PBS course from a BACB® accredited postgraduate ABA program. In addition to a traditional management structure, the Multicap PBS team employs two senior clinicians (a BCBA and registered psychologist) as practice coaches who lead detailed case reviews and provide in situ clinical mentoring.

PBS plans prepared by the team are peer reviewed using the Behaviour Intervention Plan Quality Evaluation, Version 2 (BIPQEII) (Browning-Wright et al., 2013) and presented on a standard template. The BIPQEII has formed the foundation of several studies of behaviour support plan quality and has emerged as the benchmark instrument for this purpose in Australia (McVilly et al., 2013; Wardale et al., 2018; Webber et al., 2017). The use of a standardised PBS plan template, based on the BIPQEII and mapped to the current legislative requirements for restrictive practice authorisation has been shown to increase the quality of the behavioural strategies proposed to address challenging behaviour and reduce the use of restrictive practices (Webber et al., 2017). However, this has been insufficient in effectively addressing ongoing sector issues such as staff training in non-PBS areas, rostering and recruitment of support staff, complex stakeholder management, and the capability of service teams to implement behaviour support strategies (Nankervis, 2020). In particular, there are challenges with the typical level of education of most of

the disability support workforce and the overly academic writing style used by most plan authors (Nankervis, 2020).

In addition to clinical support, Multicap has a significant focus on integration across safety, human resources, and operational functions. This integration defines Multicap's behaviour support practice approach and can be understood on four levels. Level One is a routine and annual (or during client on boarding) survey using the Behaviour Problems Inventory-Short Form (BPI-S), which highlights challenging behaviour which would be considered above the clinical norm for the individual (Rojahn et al., 2012). Individuals who score above the mean for their age range and gender are then referred for follow up with the PBS Team (see Level Two). Level Two clients have specific behavioural needs—a behaviour support clinician is allocated; a PBS Plan is prepared and close collaboration between clinical and operational services occurs to ensure the delivery of evidence-based support. Level Three is a joint initiative between clinical, operational, and human resource functions. Multicap's senior managers meet monthly to systematically review emergent risks across the organisation. This process may identify other individuals who require PBS.

Lastly, Level Four (Complex Behaviour Forum) is a joint initiative between executive leaders of operations, safety, human resources, and clinical services, including their direct reports. The Complex Behaviour Forum (CBF) seeks to support both clients and their supporters, by dedicating the most senior personnel in the organisation to an extended and collaborative problem-solving activity. Referral to the forum is based on an assessment of clinical need, usually derived from the number of incidents of challenging behaviour (>10 per month). At the discretion of the CBF chairperson, other criteria may be used such as the severity or risk presented by the behaviours. The discussions are data led, with a commitment to longitudinal and demonstrable improvement in client outcome. Clear actions are determined, and high levels of accountability are maintained in the execution of agreed responses. The CBF runs for a full day each month to consider the support needs, data, and progress, pertaining to Multicap's most complex of clients. There is a dearth of studies related to cross-departmental approaches to challenging behaviour, but one point of comparison for another complex case review panels is the Risk Reference Panel, a British panel that aims to reduce risk and recurrence of behaviour (Thomas et al., 2019). Such multisystem forums have also been used in other fields such as dementia management (Steiner et al., 2020). The CBF was initially modelled off the Victorian Government's Multiple and Complex Needs Initiative (MACNI) (Victorian Government, 2020).

During the CBF, senior operational managers, behaviour support clinicians and health and safety managers attend, to report on and consider all aspects of the CBF clients' day-to-day support, including behaviour support. They are expected to evidence the following: (1) documentation of staff training, (2) the engagement schedule of a behaviour practitioner, and (3) other standard health and wellbeing checks, e.g., Comprehensive Health Assessment Program (CHAP) and other nutrition and swallowing checklist (Lennox et al., 2007; Stewart, 2009). Behavioural data is then presented to the forum to look at the topography, frequency, duration, and intensity of the behaviours displayed by the individual.

Following this, there are discussions regarding staffing stability and risk, training gaps are identified and targeted actions for the service to follow up on are proposed. The meeting spends an average of 30 min reviewing each client's progress and these discussions are documented via meeting minutes, and actions agreed for follow up prior to the next monthly forum are documented and circulated to the relevant parties. There is a high expectation for these actions to have been completed and closed prior to the next forum appearance or sooner depending on the risk required to be addressed. This continuous review is considered necessary to correct factors such as the drift in the ability of staff to appropriately assess behaviours and implement strategies (Bonta & Andrews, 2007). At the next forum review, the actions are checked and critiqued to determine any positive impact. Each recurring appearance at the CBF commences with a review of actions, followed by data presentations; with the addition of behaviour support strategy implementation data, use of restrictive practices data, and ending with a discussion and allocation of new actions.

The organisation and implementation of Multicap's CBF approach was informed by the principles of the Victorian Government's Multiple and Complex Needs Initiative (MACNI; Victorian Government, 2020) and clinical governance. Clinical governance was originally described in efforts to reform the National Health Service in the United Kingdom (UK)—it is the term used for systems that use organisational structures and culture change to improve patient outcomes (Vassos et al., 2019). Vassos et al. (2019, p. 68) describes clinical governance as 'the implementation of a set of planned and proactive organisational approaches to ensure quality care for service users'. It is suggested that the CBF and its place in a structured PBS approach constitutes a useful model for clinical governance supporting people with challenging behaviours.

MACNI was one of the other models the CBF was based on. It was designed to provide a forum for the support of Victoria's most complex and high-risk individuals through expert advice across multiple government departments (Victorian Government, 2020). The result is enhanced service system capacity through a cross program partnership and collaborative practice within Health and Human Services (Victorian Government, 2020; Western Australian Council of Social Service, 2019). MACNI assists the individual to develop therapeutic goals and build their engagement with the appropriate services. It aims to find safe and stable housing and work on the social engagement of clients (Victorian Government, 2020). It has been described as a successful approach to supporting complex, high needs individuals and has been used as a model to develop similar approaches in other Australian states (Hamilton & Elford, 2009; Western Australian Council of Social Service, 2019). MACNI is available to people over the age of 16, who have engaged in aggressive behaviour resulting in risk of harm or actual harm to themselves or others; need intense, coordinated support; and who have at least two of the following: mental illness, acquired brain injury, intellectual disability or severe substance dependence (Victorian Government, 2020).

MACNI (and by extension, the CBF) are grounded in theoretical models like Multisystemic Therapy (MST) (Henggeler, 1999; Henggeler

et al., 1996; Littell, 2005). MST is based on the bioecological theory (Ashabi & O'Neal, 2015; Eriksson et al., 2018; Henggeler, 1999). This theory underwent significant modification over time as Bronfenbrenner refined it, initially conceptualising human development as an ecological model driven by the interaction between the person and their environment, with the environment described as systems at multiple levels (Eriksson et al., 2018). The significant lesson from this model, and others like it, is that behaviour and development is driven by the interaction of multiple complex factors. It is this acknowledgement of multifactorial causes of behaviour that drove MST (e.g., alcohol/drug problems, psychiatric issues) (Henggeler, 1999). This is also supported by the biopsychosocial model of disability which recognises that the impact of any disability is the interaction of the medical nature of the disability, the psychology of the individual, and the social context within which they act (Griffiths & Gardner, 2002). If behaviour is being driven by multiple factors, then any intervention process must be capable of addressing multiple factors, and it must be appropriate for the environment that the person lives in (i.e., interventions that involve institutionalising an individual may cease to be effective when the person returns to their home) (Burns et al., 2000).

While MST was targeted at youth offenders with mental health issues (Burns et al., 2000; Henggeler, 1999), the theoretical basis for its approach can still inform interventions like the CBF. Challenging behaviour can be the result of multiple factors. It may be driven by a functional behavioural need (e.g., accessing attention, escaping an undesired task), it may be the result of a behavioural phenotype (e.g., food seeking behaviour in Prader-Willi Syndrome), or it may be a response to environmental or physical factors (e.g., pain) (Ali et al., 2018; Emerson & Einfeld, 2011). Due to the multifactorial and complex nature of challenging behaviour, the CBF has been structured to seek input from PBS clinicians who might have insight into functional behaviour as well as specialists who can assess the individual's environment and suggest appropriate interventions, similar to other multisystem case review forums (Thomas et al., 2019).

As well as the consequences to the individual from their own challenging behaviour, client challenging behaviour can also result in increased staff injury and staff burnout (Kiely & Pankhurst, 1998; Mitchell & Hastings, 2001; Skirrow & Hatton, 2007). These risks can result in increased organisational costs due to additional and unforeseen recruitment costs due to staff turnover as a result of stress and injury and occupational health and safety claims (Gray & Muramatsu, 2013; Kozak et al., 2013; National Disability Services, 2020; National Disability Services, 2021). Given this, cross-departmental and multi-system interventions like the CBF have the capacity to not only impact client behaviour, by wider organisational outcomes of interest. In support of this, Sanders (2009) found that a multi-system approach that effectively reduced rates of physical restraint within a disability service organisation also resulted in significant cost reductions for the organisation due to a reduction in staff injury.

The aim of this study was to evaluate the effectiveness of the CBF. The study undertaken used a quasi-experimental design methodology to estimate the causal impact of the CBF on the target population (people with intellectual disability who exhibit serious challenging

behaviour). The methodology included a randomly selected group of individuals who have participated in the CBF (the intervention group) and a control group of individuals undergoing treatment as usual (i.e., PBS at Level 2 of Multicap's behaviour support practice). The control participants have not participated in the CBF but present with similar behavioural topographies and demographics (in terms of diagnosis, age, gender) to the intervention group, however the severity of the behaviours is less intense than the intervention group. Participants for this study were selected from a pool of clients receiving both accommodation services and behaviour support through Multicap.

Levels of challenging behaviour were measured at multiple phases (pre-intervention and post-intervention) to assess change in relevant outcomes over time. The organisational outcome of staff injury costs related to challenging behaviour was also considered as a secondary outcome of interest. It was hypothesised that individuals who were supported by the CBF would display greater levels of positive behaviour change than the control group supported via treatment as usual (i.e., PBS). This change would also result in a reduction in staff injury costs related to challenging behaviour.

2 | METHOD

2.1 | Participants

The total sample size was 30 individuals, split across two groups. The intervention group consisted of 15 individuals (mean age = 33.86 years, SD = 6.99 years; 11 male, 4 female) who were enlisted to receive behaviour support via the CBF and were living in Multicap supported accommodation. The control group consisted of 15 individuals (mean age = 42.53 years, SD = 13.18 years; 10 male, 5 female) who were also living in Multicap supported accommodation but were receiving Multicap Level 2 PBS services. Exclusion criteria were not living in Multicap supported accommodation and not being subject to any restrictive practice. Participants from the CBF were randomly selected to create the intervention group and control participants were selected based on the exclusion criteria due to a limited number of available participants. Ethical approval for the study was received from the University of Queensland (Approval Number: 2022/HE001993). Individual consent was not required as individual results were not being considered, just the average behavioural incident rates and staff injury rates for CBF participants against controls.

2.2 | Materials

The primary variable of interest was the number of incidents of challenging behaviour. Frequency was chosen as the preferred measure as opposed to duration or intensity for ease of comparison. These incidents included physical/verbal aggression, self-injurious behaviour, leaving supports, and property damage. These behavioural incidents were sourced from Multicap's incident management software (Riskman). This

software allows the support workers who are working with an individual to record instances of challenging behaviour, as well as the antecedents and immediate maintaining consequences. When they record the behaviour, they are required to categorise it according to a general category (e.g., physical aggression, property damage, leaving supports).

Data was also collected relating to psychological and physical injury of staff members who support the participants. This data was derived from Riskman reports as well as a workplace insurance database (WorkCover). This software allows the recording of staff injury reports with a brief description of the incident. Due to initial anonymisation of the CBF participants, this review was conducted across the entire CBF.

2.3 | Procedure

Behavioural incident data was collected for each forum participant for the three-month period before they entered the CBF (pre-intervention), their time in the CBF (intervention), and the 3 months following their time in the CBF (post-intervention). Time in CBF was calculated based on the first and last instances of an individual being discussed in the CBF; any brief breaks due to holidays or absence of clinicians were still considered to be part of the individuals time in the CBF. Control participants were matched to the CBF participants by gender and type of challenging behaviour. Age was used as a tertiary match to ensure all participants were adults. Diagnosis was not used as a selection criteria (though most of the participants had intellectual disability)—details of CBF participants are listed in Table 1, control participants are in Table 2. Comparison behavioural incidence data was gathered for the control participants matching the time period sampled for their paired CBF participant.

2.4 | Statistical analysis

All data analyses were undertaken using IBM SPSS Statistics, Version 29. Data were analysed via the calculation of descriptive statistics (means, standard deviations, frequencies, percentages) and repeated measures *t*-tests to assess for behavioural change across pre- and post-intervention phases. Due to the variable nature of the scores in the control and CBF participants, scores in both groups were transformed to standardised *z*-scores for the analysis, which allows for a consistent scale of measurement across all participants (between -3 and $+3$). This was done due to the inherent nature of the groups—the control group had fewer incidents than the CBF group since; if they had a similar incident rate, the control group participants would have been admitted to the CBF. Pre-intervention and post-intervention scores were derived for the CBF participants using the mean of the 3-month period before they entered the CBF (pre-intervention) and the 3-month period after they exited the CBF (post-intervention). The mean duration for time in the CBF for the experimental group was 10.4 months. Comparison scores were derived from the control group using matched time periods to their paired forum participant. Figure 1 presents example data from a pair of matched participants.

TABLE 1 CBF participant details.

ID	Age	Gender	Challenging behaviours	Diagnosis
CBF01	21	M	Physical aggression, self-injurious behaviour	ID, ASD
CBF02	26	M	Property damage	ASD
CBF03	28	M	Physical aggression, property damage	ID, ASD
CBF04	26	M	Physical aggression, property damage, self-injurious behaviour	ID
CBF05	32	F	Physical aggression, leaving supports	ID, ASD
CBF06	31	M	Physical aggression, self-injurious behaviour	ASD
CBF07	33	F	Physical aggression, self-injurious behaviour	ID, ASD
CBF08	35	M	Physical and verbal aggression, property damage	ID
CBF09	40	M	Physical and verbal aggression, property damage	ASD, Epilepsy, Schizophrenia
CBF10	38	F	Physical aggression, self-injurious behaviour	Chromosomal Abnormality/ASD, ID
CBF11	31	M	Physical aggression, self-injurious behaviour	Smith Magenis Syndrome, ID
CBF12	39	M	Physical and verbal aggression, property damage	ID, Schizophrenia
CBF13	46	M	Physical and verbal aggression, property damage	Huntington's Disease
CBF14	43	F	Physical aggression, self-injurious behaviour, property damage	Cri Du Chat Syndrome, ID
CBF15	39	M	Physical and verbal aggression, leaving supports	ABI

TABLE 2 Control participant details.

ID	Age	Gender	Challenging behaviours	Diagnosis
Control01	24	M	Physical and verbal aggression	Angelman Syndrome, ID
Control02	24	M	Physical and verbal aggression, self-injurious behaviour	ID, ASD, Tourette's Syndrome, ADHD, Anxiety
Control03	30	M	Property damage	ID, ADHD, Foetal Alcohol Syndrome
Control04	31	M	Physical aggression	ID, ASD
Control05	31	F	Physical and verbal aggression, self-injurious behaviour, property damage	ID, ASD, Ataxia, ADHD, Anxiety, Depression
Control06	34	M	Physical and verbal aggression, self-injurious behaviour, property damage	ID, ASD
Control07	42	F	Physical and verbal aggression, self-injurious behaviour	Agenesis of Corpus Callosum, ID, Epilepsy
Control08	43	M	Verbal aggression, self-injurious behaviour, property damage, faecal smearing	ID, ASD
Control09	44	M	Property damage	ID, ASD
Control10	45	F	Self-injurious behaviour	ID, ASD, Epilepsy
Control11	54	M	Verbal aggression, self-injurious behaviour, property damage	ID, ASD, Generalised Anxiety Disorder, Schizoaffective Disorder
Control12	57	M	Physical and verbal aggression, self-injurious behaviour	ID, ABI
Control13	57	M	Physical and verbal aggression, self-injurious behaviour	ID, ASD
Control14	59	F	Verbal aggression, self-injurious behaviour, property damage	ID, ASD
Control15	63	M	Leaving supports	Angelman Syndrome, ID

3 | RESULTS

Figure 2 presents the mean behavioural incidence z-score for the CBF and control group participants across the two phases of the study. By inspecting this graph, a group effect across time is noted, with the CBF participants showing more change in behavioural incidents across time. This was confirmed by undertaking a mixed between/within subjects ANOVA. There was a significant interaction of group and time, Wilks' Lambda = .86, $F(2, 28) = 4.66$, $p = .04$, with the CBF

showing significantly lower behavioural incident post-intervention compared to the control group. The effect size for the interaction effect was large (partial eta squared = .14).

The staff incident review covered 781 incidents involving aggressive behaviour towards staff by clients, resulting in injury to staff, between January 2018 to June 2023 (the period of data collection for this study across the 30 participants). Of those 781 incidents, 47% were related to clients who were or currently are, participants in the CBF (50 people in total, including the 15 CBF participants of this

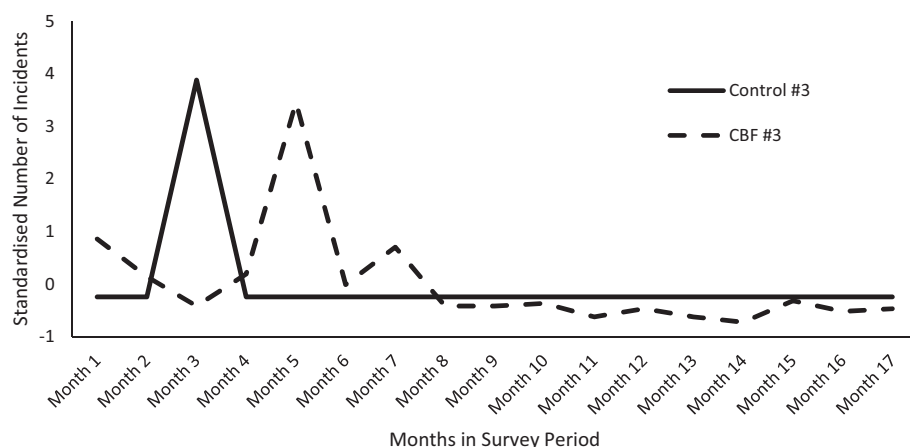


FIGURE 1 Example of data for one of the matched pairs in the study.

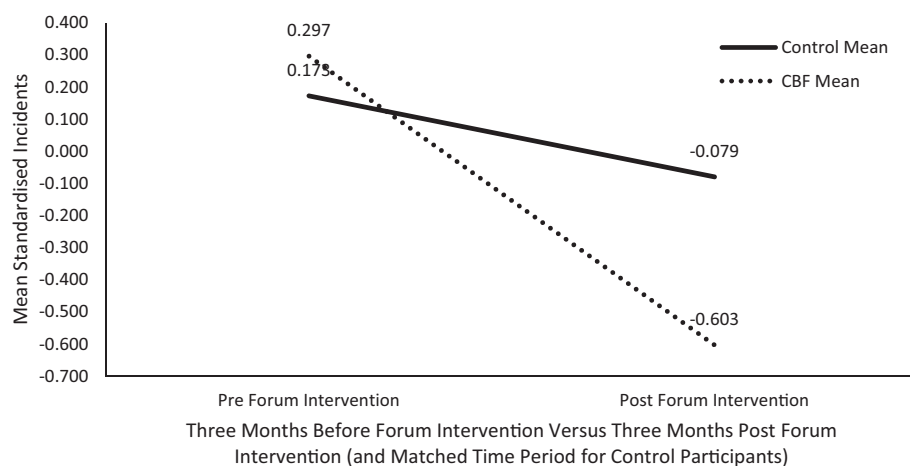


FIGURE 2 Pre and post intervention standardised means for control and CBF groups.

TABLE 3 CBF participant involvement in behavioural incidents relating to staff injury.

Incident occurrence	%
Before CBF participation (50 CBF participants)	43.1%
During CBF participation (50 CBF participants)	38.1%
After one period of CBF participation but before another (13 CBF participants)	11.7%
After CBF participation (35 CBF participants)	7.1%

study). This cohort of 50 clients makes up approximately 2.5% of Multicap's clients. The breakdown of when these 781 incidents occurred across the 50 CBF clients and in relation to the CBF is shown in Table 3. The breakdown indicates that staff injury incidents involving CBF participants (including the 15 participants of this study) decreased as a result of participation in the CBF, with a 36% decrease at post-intervention compared to pre-intervention.

4 | DISCUSSION

This study aimed to investigate the potential effectiveness of a cross-departmental approach to very complex challenging behaviours. The

findings, based on reviewing a random sample of CBF participants with matched controls who received traditional PBS, strongly suggest that the CBF intervention results in a greater degree of behavioural change. The CBF was shown to be effective for people with ID, ASD and those with cognitive issues (e.g., ABI) and neurodegenerative disorders (e.g., Huntington's disease). The data from the current study also demonstrated a reduction in challenging behaviour from standard PBS interventions, although this reduction was not statistically significant; this may be due to the small sample size. The effect for PBS intervention is in the expected and desired direction across the same 10.4 month mean time period for CBF which suggests that it is quite possible that with a larger sample size, a significant effect of PBS alone could be observable (Sullivan & Feinn, 2012). Further study with a larger sample size may assist in demonstrating the efficacy of PBS interventions in an Australian context.

The results of this study are meaningful because they provide evidence for the use of a multisystemic approach in supporting people with very complex behaviours. This study was originally driven by an internal need of the organisation to review the CBF based on the relative expense of having senior staff with significant operational responsibilities spend 1 day every month participating in the forum. In addition to that burden, there is the data collection and administration, and ad hoc participation by clinicians who are called in based on

their relationship to the clients being supported. These factors mean that the CBF constitutes a notable cost to the organisation for which it receives no additional funding. The justification for the CBF must therefore be measured in the improved outcomes it brings to the individual participants, the staff who support them and the organisation as a whole.

The reduction in behavioural incidents indicated by the results across all CBF clients (including the 15 participants of this study) suggests that this has been achieved, as does the reduction in staff injuries. An approximate annual cost for the CBF is about \$65,000 AUD (based on the salaries of attending staff, meeting for 7 h, once a month). The average cost of a staff injury Workcover claim from challenging behaviour is approximately \$42,000 AUD. Given the reduction in incidents shown in Table 3, this equates to a cost saving per year of about \$1,000,000, which implies a noteworthy financial benefit for the organisation for a small financial investment. No new staff were recruited for the CBF – it is run using only existing staff. Running this forum was a matter of allocating existing resources to achieve a cost saving. It is suggested that this model of clinical governance using a multisystemic approach may provide a model for supporting people with very challenging behaviours. Even outside of a larger corporate structure, a multisystemic approach could be adopted by PBS practitioners where they ensure that they are consulting broadly when developing PBS plans and strategies and implementing them (e.g., building consultation groups including medical practitioners, occupational therapists, support workers).

The results of the incident review lend support to the selection criteria for participants of the CBF. The disparity in proportion of staff injury incidents related to CBF clients (47%) and the proportion of the total client base who are referred to the CBF (2.5%) indicates that the individuals who are referred to the CBF are those who are presenting with the highest level of challenging behaviour and require the most complex support. It is also worth noting that 71.7% of CBF participant related claims were before or during their time in the forum. The remaining claims that occurred after a period in the CBF include participants who later came back to the forum, suggesting ongoing complexity in their profile. The nature of these incidents, particularly those that occurred before CBF intervention, can be quite extreme—an example description of one such incident is: ‘Client extremely angry chased me with big sticks and hit me with them, grabbed me, kicked me, pulled my hair with extreme force almost broke me (sic) glasses, would not let go of me. Couple of people helped me and called police, 6 x police officers arrived’. This sort of incident illustrates the type of challenging behaviour that can result in staff burnout (Kiely & Pankhurst, 1998; Mitchell & Hastings, 2001; Skirrow & Hatton, 2007). This suggests that the success of the CBF in reducing incidents may also have an impact on organisational costs.

In the Australian context, disability support funding is provided by the National Disability Insurance Scheme (NDIS) (Laragy & Fisher, 2020). The NDIS uses an individual funding model—people with disabilities apply for funding from the government and then have choice and control over how they allocate that funding for disability support services. Notwithstanding access to behaviour support

funding, the direct, individual client and transactional unit price model that the NDIS is built on makes delivery of holistic integrated approaches to supporting people with complex behaviours (like the CBF) very difficult. Further, some pricing structures such as those associated with high intensity support do not incentivise reductions in challenging behaviour, fundamentally risking the structures that facilitated the positive client behavioural outcomes demonstrated by this study. For example, ‘high intensity funding’ for operational support of a person with complex behaviour, requires that at least one behavioural incident occurs per shift (National Disability Insurance Scheme, 2022). However, the continuation of such high frequency of challenging behaviour would demonstrate behaviour support failures on multiple levels. By way of contrast, Multicap’s current response is to dedicate support across multiple functional areas to limit challenging behaviour from occurring in the first place, instead of pursuing high levels of NDIS funding, because of the anticipated and now demonstrated organisational benefits associated with fewer incidents of complex and serious challenging behaviour, for example, a reduction in staff injury and the potential reduction in organisation costs related to staff injury claims (as demonstrated by this study). What is being advocated here is that funding models do not have to dictate effective service provision—using funding wisely and creatively to ensure effective service provision can reap organisational benefits that may be worth more in the long run, not only to disability support organisations, but clients the organisation supports and their staff also.

The primary limitation of this study is the restricted sample size. This was dictated by the availability of control participants who fit the selection criteria. A larger study with more control participants would have more experimental power but, given the effect size of the behaviour change in CBF participants, this study offers appropriate initial evidence to support the effectiveness of the CBF. It is also worth noting that the samples did have different levels of challenging behaviour—this was anticipated as individuals who were engaging in high levels of challenging behaviour were likely to be referred to the CBF. Other limitations include the restriction of participants to people using Multicap accommodation services. It is unknown how the CBF model could be applied to people who are still receiving supports from a disability organisation while still living in their family home—this would be most relevant to children presenting with challenging behaviours. Ethics for this study did not cover in depth analysis of individuals—further work looking at case studies of how this type of multidisciplinary intervention may assist an individual would be one avenue of future research and would contribute to a more detailed view of multisystemic approaches. More data could also be obtained on the quality-of-life impacts for individuals receiving this type of support, specific data on restrictive practice reductions, and whether these measures had an effect on staff turnover. Due to this study being conducted within a corporate environment, this sort of data was not collected during the usual course of business—a more experimental approach which did focus on this type of data collection would provide valuable insight. It is also acknowledged that data collection in a corporate environment may not always have the rigour of experimental settings, but the strength of the relationships detected not

only suggest the value of multisystemic approaches, but the quality of the data collected also. The final limitation of this study also relates to the setting—the use of a multisystemic approach in a corporate environment means that it is difficult to establish if there was a specific component of the CBF approach that yielded the most benefit to individuals (i.e., the interdisciplinary collaborative work or the intensive oversight and accountability components).

This presents one possibility for future research—the exploration of how clinical governance models using a cross-departmental approach (like the CBF) may function in different settings (e.g., family home, aged care). Other possibilities for future research have already been proposed as the organisation used in this study (Multicap) has a Complex Health Forum and a Complex Mental Health Forum. Both of these forums use a similar cohort of representatives from across the organisation and represent opportunities to explore if this cross-departmental model of clinical governance can be effective in other conditions. A review of the Complex Health Forum is currently in progress. Another area of future research would be to examine staff burnout as an outcome of interest alongside behavioural incidents and staff injury. This paper has alluded to a link between challenging behaviour, staff burnout and staff injury (Mitchell & Hastings, 2001; Skirrow & Hatton, 2007), however there is limited contemporary PBS-related intervention research that looks at staff burnout as an outcome of interest.

The CBF is part of an ongoing and strategic response to people with complex needs. This is defined by attention to advanced clinical skills, timely and data driven responses and integration between multiple functions of the organisation. This does however present some challenges when supporting clients who exercise their right to choose an external clinical (behaviour support) provider. This typically occurs where a client or their representative engages a third-party clinical provider directly, via the use of their NDIS funds. In these situations, Multicap may not be aware of the capability of the clinician, or the organisational approach to their training and development. Multicap is also unable to direct the activity of these clinicians, risking poor clinical response, ill-considered behavioural interventions, and poor collaboration between the clinician and Multicap personnel. This lack of a unified approach has presented a real and significant risk to client safety. The structure of the NDIS allows for greater choice and control for people with disabilities, but it is worth noting that this may create issues for organising services across multiple systems which represents a challenge for coordinating responses to complex and challenging behaviour.

Whilst large scale policy initiatives such as the NDIS in Australia can facilitate broad and generalised improvement, they cannot match the standards achieved by carefully structured, long term organisational strategy. In this regard, standards across the sector may generally improve, but the unique and highly complex needs of an important but vulnerable group may continue to be overlooked. It is proposed that one method for organisations seeking to improve their clinical outcomes is to adopt governance measures like the CBF. This paper provides evidence that this type of structure can be an effective addition to standard PBS approaches when a heightened response is required due to very complex behaviour.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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